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Inventor(s)	JUNG; Gwangchae et al.

ELECTRONIC DEVICE COMPRISING FLEXIBLE DISPLAY, AND METHOD FOR CONTROLLING SAME

Abstract

An electronic device may include: a housing; at least one sensor; a flexible display having at least part exposed to the outside through the housing; and at least one processor operatively connected to the at least one sensor and the flexible display, wherein the at least one processor may identify through the at least one sensor whether at least a part of the flexible display is covered while the flexible display is slid in with respect to the housing, perform a sliding-out operation of the flexible display on the basis of an event occurring while at least a part of the flexible display is covered, and display information related to the event, in an area expanded by the sliding-out operation.

Inventors: **JUNG; Gwangchae (Suwon-si, KR), KWON; Kijin (Suwon-si, KR), KIM; Kisung (Suwon-si, KR), PARK; Minhoo (Suwon-si, KR), PARK; Sangil (Suwon-si, KR), LEE; Wooseok (Suwon-si, KR), CHUNG; Chongyoon (Suwon-si, KR), HA; Dohyung (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/017131 designating the United States, filed on Oct. 31, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0163841 filed on Nov. 30, 2022, and Korean Patent Application 10-2022-0176435 filed on Dec. 16, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD
[0002] Certain example embodiments may relate to an electronic device including a flexible display and/or a method for controlling the same.
BACKGROUND ART
[0003] As the demand for mobile communication increases, or as the degree of integration of electronic devices increases, the portability of electronic devices such as mobile communication terminals may be increased, and better convenience may be provided in use of multimedia functions. For example, as touchscreen-integrated displays replace traditional mechanical (button-type) keypads, electronic devices may come more compact while functioning as an input device. For example, as the mechanical keypad may be omitted from the electronic device, portability of the electronic device may be improved. As the display area may be expanded to the area which used to be occupied by the mechanical keypad, the electronic device may provide a larger screen while remaining in the same size and weight as when it has the mechanical keypad.
[0004] Use of an electronic device with a larger screen may give more convenience in, e.g., web browsing or multimedia playing. A larger display may be adopted to output a larger screen. However, this way may be limited by the portability of the electronic device. According to an embodiment, a display using organic light emitting diodes may secure the portability of the electronic device while providing a larger screen. For example, a display using, or equipped with, organic light emitting diodes may implement a stable operation even if it is made quite thin, so that the display may be applied to an electronic device in a foldable, bendable or rollable form.
SUMMARY
[0005] According to an example embodiment, an electronic device may comprise a housing, at least one sensor, a flexible display at least partially visible to an outside through the housing, memory storing at least one instruction, and at least one processor, comprising processing circuitry, operatively connected directly or indirectly to the at least one sensor, the flexible display and the memory.
[0006] According to an example embodiment, the at least one processor may identify whether a portion of the flexible display is covered through the at least one sensor in a state in which the flexible display slides in with respect to the housing.
[0007] According to an example embodiment, the at least one processor may perform a slide-out operation of the flexible display based on an occurrence of an event in the state in which the

portion of the flexible display is covered.
[0008] According to an example embodiment, the at least one processor may display information related to the event in an area exposed by the slide-out operation, wherein the exposed area corresponds to the covered portion.
[0009] According to an example embodiment, a method for controlling an electronic device may comprise identifying whether a portion of a flexible display is covered through at least one sensor in a state in which the flexible display slides in with respect to a housing.
[0010] According to an example embodiment, the method for controlling the electronic device may comprise sliding out the flexible display based on an occurrence of an event in the state in which the portion of the flexible display is covered.
[0011] According to an example embodiment, the method for controlling the electronic device may comprise displaying information related to the event in an area exposed by the slide-out operation, wherein the exposed area corresponds to the covered portion.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0012] FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an example embodiment;
[0013] FIG. **2** is a view illustrating a state in which a second display area of a display is received in a housing according to an example embodiment.
[0014] FIG. **3** is a view illustrating a state in which a second display area of a display is exposed to the outside of a housing according to an example embodiment;
[0015] FIG. **4** is an exploded perspective view illustrating an electronic device according to an example embodiment;
[0016] FIG. **5**A is a cross-sectional view taken along line A-A′ of FIG. **2** according to an example embodiment;
[0017] FIG. **5**B is a cross-sectional view taken along line B-B′ of FIG. **3** according to an example embodiment;
[0018] FIG. **6** is a flowchart illustrating an extending operation when an event occurs in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;
[0019] FIG. **7** is a view illustrating a use example when an electronic device is mounted on a flip cover according to an example embodiment;
[0020] FIG. **8**A is a view illustrating an operation of extending a flexible display of an electronic device in a vertical direction in a state in which at least a portion thereof is covered according to an example embodiment;
[0021] FIG. **8**B is a view illustrating an operation of extending a flexible display of an electronic device in a horizontal direction in a state in which at least a portion thereof is covered according to an example embodiment;
[0022] FIG. **9**A is a view illustrating an extending operation based on an application where an event occurs, in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;
[0023] FIG. **9**B is a view illustrating an extending operation based on an application where an event occurs, in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;
[0024] FIG. **10** is a view illustrating a stepwise extending operation when an event occurs in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;

[0025] FIG. **11**A is a view illustrating an operation in which a screen configuration is varied during stepwise extension in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;
[0026] FIG. **11**B is a view illustrating an operation in which a screen configuration is varied during stepwise extension in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment;
[0027] FIG. **12** is a view illustrating an operation of obtaining an image based on an aspect ratio of an extended area when a camera application is executed in a state in which at least a portion thereof a flexible display of an electronic device is covered according to an example embodiment;
[0028] FIG. **13**A is a view illustrating a configuration screen for receiving a user input for setting to display a widget when a flexible display is extended based on an occurrence of an event in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment; and
[0029] FIG. **13**B is a view illustrating a widget displayed based on a degree of extension of a flexible display in a state in which at least a portion of a flexible display of an electronic device is covered according to an example embodiment.

DETAILED DESCRIPTION

[0030] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with at least one of an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In an embodiment, at least one (e.g., the connecting terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. According to an embodiment, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be integrated into a single component (e.g., the display module **160**).
[0031] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be configured to use lower power than the main processor **121** or to be specified for a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
[0032] The auxiliary processor **123** may control at least some of functions or states related to at

least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. The artificial intelligence model may be generated via machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0033] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0034] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0035] The input module **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, keys (e.g., buttons), or a digital pen (e.g., a stylus pen).

[0036] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0037] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display **160** may include a touch sensor configured to detect a touch, or a pressure sensor configured to measure the intensity of a force generated by the touch.

[0038] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0039] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an accelerometer, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0040] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0041] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0042] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0043] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0044] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0045] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0046] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via a first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0047] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0048] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module **197** may include one antenna including a radiator formed of a conductor or conductive pattern formed on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., an antenna array). In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected from the plurality of antennas by, e.g., the communication module **190**. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **197**.

[0049] According to an embodiment, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0050] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0051] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. The external electronic devices **102** or **104** each may be a device of the same or a different type from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing

or mobile edge computing. In another embodiment, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or health-care) based on 5G communication technology or IoT-related technology.
[0052] FIG. **2** is a view illustrating a state in which a second display area of a display is received in a housing according to an embodiment. FIG. **3** is a view illustrating a state in which a second display area of a display is exposed to the outside of a housing according to an embodiment.
[0053] FIGS. **2** and **3** illustrate a structure in which the display **203** (e.g., flexible display or rollable display) is extended in the length direction (e.g., +Y direction) when the electronic device **101** is viewed from the front. However, the extending direction of the display **203** is not limited to one direction (e.g., +Y direction). For example, the extending direction of the display **203** may be changed in design to be extendable in the upper direction (+Y direction), right direction (e.g., +X direction), left direction (e.g., −X direction), and/or lower direction (e.g., −Y direction).
[0054] The state shown in FIG. **2** may be referred to as a closed state of the electronic device **101** or housing **210** and a slide-in state of the display **203**.
[0055] The state shown in FIG. **3** may be referred to as an opened state of the electronic device **101** or housing **210** and a slide-out state of the display **203**.
[0056] Referring to FIGS. **2** and **3**, the electronic device **101** may include a housing **210**. The housing **210** may include a first housing **202** and a second housing **202** disposed to be movable relative to the first housing **201**. According to an embodiment, the electronic device **101** may be interpreted as having a structure in which the first housing **201** is disposed to be slidable with respect to the second housing **202**. According to an embodiment, the second housing **202** may be disposed to perform reciprocating motion by a predetermined distance in a predetermined direction with respect to the first housing **201**, for example, a direction indicated by an arrow {circle around (**1**)}.
[0057] According to an embodiment, the second housing **202** may be referred to as a slide portion or a slide housing, and may be movable relative to the first housing **201**. According to an embodiment, the second housing **202** may receive various electrical/electronic components, such as a circuit board or a battery.
[0058] According to an embodiment, the first housing **202** may have, disposed therein, a motor, a speaker, a sim socket, and/or a sub circuit board electrically connected with a main circuit board. The second housing **201** may receive a main circuit board on which electric components, such as an application processor (AP) and a communication processor (CP) are mounted.
[0059] According to an embodiment, the first housing **201** may include a first cover member **211** (e.g., a main case). The first cover member **211** may include a 1-1th sidewall **211***a*, a 1-2th sidewall **211***b* extending from the 1-1th sidewall **211***a*, and a 1-3th sidewall **211***c* extending from the 1-1th sidewall **211***a* and substantially parallel to the 1-2th sidewall **211***b*. According to an embodiment, the 1-2th sidewall **211***b* and the 1-3th sidewall **211***c* may be formed substantially perpendicular to the 1-1th sidewall **211***a*.
[0060] According to an embodiment, the 1-1th sidewall **211***a*, 1-2th sidewall **211***b*, and 1-3th sidewall **211***c* of the first cover member **211** may be formed to have a side opening (e.g., front opening) to receive (or surround) at least a portion of the second housing **202**. For example, at least a portion of the second housing **202** may be surrounded by the first housing **201** and be slid in the direction parallel to the first surface (e.g., the first surface F**1** of FIG. **4**), e.g., arrow {circle around (**1**)} direction, while being guided by the first housing **201**. According to an embodiment, the 1-1th sidewall **211***a*, the 1-2th sidewall **211***b*, and/or the 1-3th sidewall **211***c* of the first cover member **211** may be integrally formed. According to an embodiment, the 1-1th sidewall **211***a*, the 1-2th sidewall **211***b*, and/or the 1-3th sidewall **211***c* of the first cover member **211** may be formed as

separate structures and be combined or assembled.

[0061] According to an embodiment, the first cover member **211** may be formed to surround at least a portion of the display **203**. For example, at least a portion of the display **203** may be formed to be surrounded by the 1-1th sidewall **211***a*, the 1-2th sidewall **211***b*, and/or the 1-3th sidewall **211***c* of the first cover member **211**.

[0062] According to an embodiment, the second housing **202** may include a second cover member **221** (e.g., a slide plate). The second cover member **221** may have a plate shape and include a first surface (e.g., the first surface F**1** of FIG. **4**) supporting internal components. For example, the second cover member **221** may support at least a portion of the display **203** (e.g., the first display area A**1**). According to an embodiment, the second cover member **221** may be referred to as a front cover.

[0063] According to an embodiment, the second cover member **221** may include a 2-1th sidewall **221***a*, a 2-2th sidewall **221***b* extending from the 2-1th sidewall **221***a*, and a 2-3th sidewall **221***c* extending from the 2-1th sidewall **221***a* and substantially parallel to the 2-2th sidewall **221***b*. According to an embodiment, the 2-2th sidewall **221***b* and the 2-3th sidewall **221***c* may be formed substantially perpendicular to the 2-1th sidewall **221***a*.

[0064] According to an embodiment, as the second housing **202** moves in a first direction (e.g., direction {circle around (**1**)}) parallel to the 1-2th sidewall **211***b* or the 1-2th sidewall **211***c*, the housing **210** may form an opened state and a closed state. In the closed state, the second housing **202** may be positioned at a first distance from the 1-1th sidewall **211***a* and, in the opened state, the second housing **202** may be moved to be positioned at a second distance larger than the first distance from the 1-1th sidewall **211***a*. In some embodiments, in the closed state, the first housing **201** may surround a portion of the 2-1th side wall **221***a*.

[0065] According to an embodiment, the electronic device **101** may include a display **203**, a key input device **245**, a connector hole **243**, audio modules **247***a* and **247***b*, or camera modules **249***a* and **249***b*. According to an embodiment, the electronic device **101** may further include an indicator (e.g., a light emitting diode (LED) device) or various sensor modules.

[0066] According to an embodiment, the display **203** may include a first display area A**1** configured to remain exposed and a second display area A**2** configured to be exposed to the outside of the electronic device **101** based on the slide of the second housing **202**. According to an embodiment, the first display area A**1** may be disposed on the second housing **202**. For example, the first display area A**1** may be disposed on the second cover member **221** of the second housing **202**. According to an embodiment, the second display area A**2** may extend from the first display area A**1**, and as the second housing **202** slides relative to the first housing **201**, the second display area A**2** may be received in the first housing **201** (e.g., the slide-in state) or be visually exposed to the outside of the electronic device **101** (e.g., the slide-out state).

[0067] According to an embodiment, the second display area A**2** may be received in the space positioned inside the first housing **201** or exposed to the outside of the electronic device while being substantially guided by one area (e.g., the curved surface **213***a* of FIG. **4**) of the first housing **201**. According to an embodiment, the second display area A**2** may move based on a slide of the second housing **202** in the first direction (e.g., the direction indicated by the arrow {circle around (**1**)}. For example, while the second housing **202** slides, a portion of the second display area A**2** may be deformed into a curved shape in a position corresponding to the curved surface **213***a* of the first housing **201**.

[0068] According to an embodiment, when viewed from above the second cover member **221** (e.g., front cover), if the electronic device **210** changes from the closed state to opened state (e.g., if the second housing **202** slides to extend from the first housing **201**), the second display area A**2** may be gradually exposed to the outside of the first housing **201** and, together with the first display area A**1**, form a substantially flat surface. According to an embodiment, the display **203** may be coupled with or disposed adjacent to a touch detection circuit, a pressure sensor capable of measuring the strength (pressure) of touches, and/or a digitizer for detecting a magnetic field-type stylus pen. According to an embodiment, irrespective of the closed state or opened state of the housing **210**, the exposed portion of the second display area A**2** may be positioned on a portion (e.g., the curved surface **213***a* of FIG. **4**) of the first housing, and a portion of the second display area A**2** may remain in the curved shape in the position corresponding to the curved surface **213***a*.

[0069] According to an embodiment, the key input device **245** may be positioned in one area of the first housing **201**. Depending on the appearance and the state of use, the electronic device **101** may be designed to omit the illustrated key input device **245** or to include additional key input device(s). According to an embodiment, the electronic device **101** may include a key input device (not shown), e.g., a home key button or a touchpad disposed around the home key button. According to an embodiment, at least a portion of the key input device **245** may be disposed on the 1-1th sidewall **211***a*, the 1-2th sidewall **211***b*, or the 1-3th sidewall **211***c* of the first housing **201**.

[0070] According to an embodiment, the connector hole **243** may be omitted or may accommodate a connector (e.g., a universal serial bus (USB) connector) for transmitting and receiving power and/or data with an external electronic device. According to an embodiment (not shown), the electronic device **101** may include a plurality of connector holes **243**, and some of the plurality of connector holes **243** may function as connector holes for transmitting/receiving audio signals with an external electronic device. In the illustrated embodiment, the connector hole **243** is disposed in the second housing **202**, but is not limited thereto. For example, the connector hole **243** or a connector hole not shown may be disposed in the first housing **201**.

[0071] According to an embodiment, the audio modules **247***a* and **247***b* may include at least one speaker hole **247***a* or at least one microphone hole **247***b*. One of the speaker holes **247***a* may be provided as a receiver hole for voice calls, and the other may be provided as an external speaker hole. The electronic device **101** may include a microphone for obtaining sound. The microphone may obtain external sound of the electronic device **101** through the microphone hole **247***b*. According to an embodiment, the electronic device **101** may include a plurality of microphones to detect the direction of sound. According to an embodiment, the electronic device **101** may include an audio module in which the speaker hole **247***a* and the microphone hole **247***b* are implemented as one hole or may include a speaker without the speaker hole **247***a* (e.g., a piezo speaker).

[0072] According to an embodiment, the camera modules **249***a* and **249***b* may include a first camera module **249***a* (e.g., a front camera) and a second camera module **249***b* (e.g., a rear camera) (e.g., the second camera module **249***b* of FIGS. **5**A and **5**B). According to an embodiment, the electronic device **101** may include at least one of a wide-angle camera, a telephoto camera, or a close-up camera. According to an embodiment, the electronic device **200** may measure the distance to the subject by including an infrared projector and/or an infrared receiver. The camera modules **249***a* and **249***b* may include one or more lenses, an image sensor, and/or an image signal processor. The first camera module **249***a* may be disposed to face in the same direction as the display **203**. For example, the first camera module **249***a* may be disposed in an area around the first display area A**1** or overlapping the display **203**. When disposed in the area overlapping the display **203**, the first camera module **249***a* may capture the subject through the display **203**. According to an embodiment, the first camera module **249***a* may include an under display camera (UDC) that has a screen display area (e.g., the first display area A**1**) that may not be visually exposed but hidden. According to an embodiment, the second camera module **249***b* may capture the subject in a direction opposite to the first display area A**1**. According to an embodiment, the first camera module **249***a* and/or the second camera module **249***b* may be disposed on the second housing **202**.

[0073] According to an embodiment, an indicator (not shown) of the electronic device **101** may be disposed on the first housing **201** or the second housing **202**, and the indicator may include a light emitting diode to provide state information about the electronic device **101** as a visual signal. The sensor module (not shown) of the electronic device **101** may produce an electrical signal or data value corresponding to the internal operation state or external environment state of the electronic

device. The sensor module may include, for example, a proximity sensor, a fingerprint sensor, or a biometric sensor (e.g., an iris/face recognition sensor or a heartrate monitor (HRM) sensor). According to another embodiment, the sensor module may further include, e.g., at least one of a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
[0074] FIG. **4** is an exploded perspective view illustrating an electronic device according to an embodiment.
[0075] FIG. **5**A is a cross-sectional view taken along line A-A′ of FIG. **2** according to an embodiment.
[0076] FIG. **5**B is a cross-sectional view taken along line B-B′ of FIG. **3** according to an embodiment.
[0077] Referring to FIGS. **4**, **5**A, and/or **5**B, an electronic device **101** may include a first housing **201**, a second housing **202**, a display assembly **230**, and a driving structure **240**. The configuration of the first housing **201**, the second housing **202**, and the display assembly **230** of FIGS. **4**, **5**A, and/or **5**B may be identical in whole or part to the configuration of the first housing **201**, the second housing **202**, and the display **203** of FIGS. **2** and/or **3**.
[0078] According to an embodiment, the first housing **201** may include a first cover member **211** (e.g., the first cover member **211** of FIGS. **2** and **3**), a frame **213**, and a first rear plate **215**.
[0079] According to an embodiment, the first cover member **211** may accommodate at least a portion of the frame **213** and accommodate a component (e.g., battery **289**) positioned in the frame **213**. According to an embodiment, the first cover member **211** may be formed to surround at least a portion of the second housing **202**. According to an embodiment, the second circuit board **249** receiving the electronic component (e.g., the processor **120** and/or the memory **130** of FIG. **1**) may be connected to the first cover member **211**.
[0080] According to an embodiment, the frame **213** may be connected, directly or indirectly, to the first cover member **211**. For example, the frame **213** may be connected to the first cover member **211**. The second housing **202** is movable relative to the first cover member **211** and/or the frame **213**. According to an embodiment, the frame **213** may accommodate the battery **289**. According to an embodiment, the frame **213** may include a curved portion **213***a* facing the display assembly **230**.
[0081] According to an embodiment, the first rear plate **215** may substantially form at least a portion of the exterior of the first housing **201** or the electronic device **101**. For example, the first rear plate **215** may be coupled to the outer surface of the first cover member **221**. According to an embodiment, the first rear plate **215** may provide a decorative effect on the exterior of the electronic device **101**. The first rear plate **215** may be formed of at least one of metal, glass, synthetic resin, or ceramic.
[0082] According to an embodiment, the second housing **202** may include a second cover member **221** (e.g., the second cover member **221** of FIGS. **2** and **3**), a rear cover **223**, and a second rear plate **225**.
[0083] According to an embodiment, the second cover member **221** may be connected to the first housing **201** through the guide rail **250** and, while being guided by the guide rail **250**, reciprocate linearly in one direction (e.g., the direction of arrow {circle around (**1**)} in FIG. **3**).
[0084] According to an embodiment, the second cover member **221** may support at least a portion of the display **203**. For example, the second cover member **221** may include a first surface F**1**. The first display area A**1** of the display **203** may be substantially positioned on the first surface F**1** to maintain a flat panel shape. According to an embodiment, the second cover member **221** may be formed of a metal material and/or a non-metal (e.g., polymer) material. According to an embodiment, the first circuit board **248** accommodating the electronic component (e.g., the processor **120** and/or the memory **130** of FIG. **1**) may be connected to the second cover member **221**.

[0085] According to an embodiment, the rear cover **223** may protect a component (e.g., the first circuit board **248**) positioned on the second cover member **221**. For example, the rear cover **223** may be connected to the second cover member **221** and may be formed to surround at least a portion of the first circuit board **248**. According to an embodiment, the rear cover **223** may include an antenna pattern for communicating with an external electronic device. For example, the rear cover **223** may include a laser direct structuring (LDS) antenna.
[0086] According to an embodiment, the second rear plate **225** may substantially form at least a portion of the exterior of the second housing **202** or the electronic device **101**. For example, the second rear plate **225** may be coupled to the outer surface of the second cover member **221**. According to an embodiment, the second rear plate **225** may provide a decorative effect on the exterior of the electronic device **101**. The second rear plate **215** may be formed of at least one of metal, glass, synthetic resin, or ceramic.
[0087] According to an embodiment, the display assembly **230** may include a display **231** (e.g., the display **203** of FIGS. **2** and/or **3**) and a multi-bar structure **232** supporting the display **203**. According to an embodiment, the display **231** may be referred to as a flexible display, a foldable display, and/or a rollable display.
[0088] According to an embodiment, the multi-bar structure **232** may be connected to or attached to at least a portion (e.g., the second display area A**2**) of the display **231**. According to an embodiment, as the second housing **202** slides, the multi-bar structure **232** may move with respect to the first housing **201**. In the closed state of the electronic device **101** (e.g., FIG. **2**), the multi-bar structure **232** may be mostly received in the first housing **201** and may be positioned between the first cover member **211** and the second cover member **221**. According to an embodiment, at least a portion of the multi-bar structure **232** may move corresponding to the curved surface **213***a* positioned at the edge of the frame **213**. According to an embodiment, the multi-bar structure **232** may be referred to as a display supporting member or supporting structure and may be in the form of one elastic plate.
[0089] According to an embodiment, the driving structure **240** may move the second housing **202** relative to the first housing **201**. For example, the drive structure **240** may include a motor **241** configured to generate a driving force for sliding the housing **201** and **202**. The driving structure **240** may include a gear (e.g., a pinion) connected to the motor **241** and a rack **242** configured to mesh with the gear.
[0090] According to an embodiment, the housing in which the rack **242** is positioned and the housing in which the motor **241** is positioned may be different. According to an embodiment, the motor **241** may be connected to the second housing **202**. The rack **242** may be connected to the first housing **201**. According to another embodiment, the motor **241** may be connected to the first housing **201**. The rack **242** may be connected to the second housing **202**.
[0091] According to an embodiment, the first housing **201** may receive the first circuit p board **248** (e.g., a main board). According to an embodiment, the processor, memory, and/or interface may be mounted on the first circuit board **248**. The processor may include one or more of, e.g., a central processing unit, an application processor, a graphic processing device, an image signal processing, a sensor hub processor, or a communication processor. According to an embodiment, the first circuit board **248** may include a flexible printed circuit board type radio frequency cable (FRC). The first circuit board **248** may be disposed on at least a portion of the second cover member **221** and may be electrically connected to the antenna module and the communication module.
[0092] According to an embodiment, the memory may include, e.g., a volatile or non-volatile memory.
[0093] According to an embodiment, the interface may include, e.g., a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, and/or an audio interface. The interface may electrically or physically connect, e.g., the electronic device **101** with an external electronic device and may include a USB connector, an SD

card/multimedia card (MMC) connector, or an audio connector.

[0094] According to an embodiment, the electronic device **101** may include a second circuit board **249** (e.g., a sub circuit board) spaced apart from the first circuit board **248** (e.g., a main circuit board) in the first housing **201**. The second circuit board **249** may be electrically connected to the first circuit board **248** through a connection flexible board. The second circuit board **249** may be electrically connected with electric components disposed in an end area of the electronic device **101**, such as the battery **289** or a speaker and/or a sim socket, and may transfer signals and power. According to an embodiment, the second circuit board **249** may receive a wireless charging antenna (e.g., coil). For example, the battery **289** may receive power from an external electronic device through the wireless charging antenna. As another example, the battery **289** may transfer power to the external electronic device by the wireless charging antenna.

[0095] According to an embodiment, the battery **289** may be a device for supplying power to at least one component of the electronic device **101**. The battery **189** may include a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell. The battery **289** may be integrally or detachably disposed inside the electronic device **101**. According to an embodiment, the battery **289** may be formed of a single embedded battery or may include a plurality of removable batteries. According to an embodiment, the battery **289** may be positioned in a frame **213**, and the battery **289** may be slid along with the frame **213**.

[0096] According to an embodiment, the guide rail **250** may guide the movement of the multi-bar structure **232**. For example, the multi-bar structure **232** may slide along the slit **251** formed in the guide rail **250**. According to an embodiment, the guide rail **250** may be connected to the first housing **201**. For example, the guide rail **250** may be connected to the first cover member **211** and/or the frame **213**. According to an embodiment, the slit **251** may be referred to as a groove or recess formed in the inner surface of the guide rail **250**.

[0097] According to an embodiment, the guide rail **250** may provide pressure to the multi-bar structure **233** based on the driving of the motor **241**.

[0098] According to an embodiment, when the electronic device **101** changes from the closed state to opened state, the inner portion **252** of the guide rail **250** may provide pressure to the multi-bar structure **232**. The multi-bar structure **232** receiving the pressure may be moved along the slit **251** of the guide rail **250**, and the second housing **202** may be changed from the slide-in state to slide-out state with respect to the first housing **201**. At least a portion of the display assembly **230** accommodated between the first cover member **211** and the frame **213** may be extended to the front surface.

[0099] According to an embodiment, when the electronic device **101** changes from the opened state to closed state, an outer portion **253** of the guide rail **250** may provide pressure to the bent multi-bar structure **232**. The multi-bar structure **232** receiving the pressure may be moved along the slit **251** of the guide rail **250**, and the second housing **202** may be changed from the slide-out state to slide-in state with respect to the first housing **201**. At least a portion of the display assembly **230** may be accommodated between the first cover member **211** and the frame **213**.

[0100] Referring to FIG. **5**A, in the closed state of the electronic device **101**, at least a portion of the second housing **202** may be disposed to be received in the first housing **201**. As the second housing **202** is disposed to be received in the first housing **201**, the overall volume of the electronic device **101** may be reduced. According to an embodiment, when the second housing **202** is received in the first housing **201**, the size of the visually exposed display **231** may be minimized or reduced. For example, if the second housing **202** is fully received in the first housing **201**, the first display area A**1** of the display **231** may be visually exposed, and the second display area A**2** may not be visually exposed. At least a portion of the second display area A**2** may be disposed between the battery **289** and the rear plate **215** and **225**.

[0101] Referring to FIG. **5**B, in the opened state of the electronic device **101**, at least a portion of the second housing **202** may protrude from the first housing **201**. As the second housing **202** is disposed to protrude from the first housing **201**, the overall volume of the electronic device **101** may be increased. According to an embodiment, if the second housing **202** protrudes from the first housing **201**, at least a portion of the second display area A**2** of the display **231**, together with the first display area A**1**, may be visually exposed to the outside of the electronic device **101**.

[0102] FIG. **6** is a flowchart illustrating an extending operation when an event occurs in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.

[0103] Referring to FIG. **6**, in operation **610**, the electronic device (e.g., the electronic device **101** of FIG. **1**, the processor **120** of FIG. **1**) may identify whether at least a portion of the flexible display is covered through at least one sensor (e.g., the sensor module **176** of FIG. **1**) in a state in which the display module **160** of FIG. **1** slides in with respect to the housing.

[0104] According to an embodiment, the electronic device may identify that at least a portion of the flexible device is covered based on identifying that the flip cover is closed through at least one sensor (e.g., a proximity sensor, an illumination sensor, and/or a hall IC sensor). According to an embodiment, the flip cover may be fastened to a portion of the housing of the electronic device to cover at least a portion of the surface of the flexible display. According to an embodiment, even when at least a portion of the flexible display is covered by not only the flip cover but also a pocket-shaped cover, the user's hand and/or an object, the electronic device may identify that at least a portion of the flexible device is covered through at least one sensor.

[0105] According to an embodiment, when the flip cover is fastened to the electronic device, an example of using the electronic device is described in more detail with reference to FIG. **7**.

[0106] According to an embodiment, in operation **620**, the electronic device may perform a slide-out operation of the flexible display based on an occurrence of an event in a state in which at least a portion of the flexible display is covered.

[0107] According to an embodiment, the electronic device may perform a slide-out operation of the flexible display stepwise or continuously.

[0108] For example, the electronic device may be extended by a second size by a user input in a state in which the flexible display is extended by a first size, and be extended by a third size in a state in which the flexible display is extended by the second size.

[0109] According to an embodiment, when a user input for extending the flexible display is continuously received (e.g., when a physical key for extension is long pressed), the electronic device may continuously extend the flexible display.

[0110] According to an embodiment, an operation of extending the flexible display is described below with reference to FIGS. **8**A and **8**B.

[0111] According to an embodiment, in operation **630**, the electronic device may display information related to the event in the area extended by the slide-out operation.

[0112] According to an embodiment, the electronic device may determine an extension size of the flexible display based on an application corresponding to the event. According to an embodiment, the extension size of the flexible display according to the application may be set by the manufacturer and/or the user. For example, the flexible display may be configured to be extended by a first size for a music player application, by a second size larger than the first size for a message application and/or a first SNS application, and by a third size larger than the second size for a video player application, a second SNS application, or a camera application.

[0113] According to an embodiment, an operation of extending the flexible display based on the application corresponding to the generated event is described below with reference to FIGS. **9**A and **9**B.

[0114] According to an embodiment, the electronic device may gradually extend the flexible display based on a user input after the event occurs.

[0115] According to an embodiment, the electronic device may perform a slide-out operation of the flexible display so that the flexible display is extended by the first size based on the occurrence of the event. According to an embodiment, the electronic device may display a color corresponding to an application corresponding to the event in an extended area having the first size. According to an embodiment, the color corresponding to the application may be a main color of an application icon.

[0116] According to an embodiment, the electronic device may perform the slide-out operation of the flexible display so that the flexible display is extended by the second size larger than the first size based on receiving the user input for display extension in a state in which the flexible display is extended by the first size. For example, the electronic device may perform the slide-out operation of the flexible display so that the flexible display is extended by the second size larger than the first size when the extended area of the flexible display is touched or a physical key for extending the flexible display is pressed in a state in which the flexible display is extended by the first size.

[0117] According to an embodiment, the electronic device may display information related to the event corresponding to the second size in the extended area of the second size. According to an embodiment, the electronic device may display a user interface capable of controlling content and/or summary of content related to the event in the extended area having the second size.

[0118] According to an embodiment, the electronic device may divide the extended area of the second size into a left area and a right area, and display different information related to the event in the divided areas. According to an embodiment, when the second size is larger than or equal to a set value and the event-related application includes a function of displaying an image and/or a video, the electronic device may divide the extended area having the second size into a left area and a right area, display a user interface capable of controlling content and/or summary of content related to the event in one area, and display an image and/or video (e.g., a visual interface (VI)) of the application in the other area.

[0119] According to an embodiment, if the flexible display is configured to display a widget when its size is the second size or more, the electronic device may display a widget in one of the divided areas and display a summary of content related to the event, a user interface capable of controlling content and/or an image (or video) of the application in the other area.

[0120] According to an embodiment, the electronic device may perform a slide-out operation of the flexible display so that the flexible display is extended by the third size larger than the second size based on receiving a user input in a state in which the flexible display is extended by the second size. For example, the electronic device may perform the slide-out operation of the flexible display so that the flexible display is extended by the third size larger than the second size when the extended area of the flexible display is touched or a physical key for extending the flexible display is pressed in a state in which the flexible display is extended by the second size.

[0121] According to an embodiment, the electronic device may display information related to the event corresponding to the third size in the extended area having the third size. According to an embodiment, the electronic device may display a user interface capable of controlling content and/or at least a portion of the entire content related to the event in the extended area having the third size.

[0122] According to an embodiment, the electronic device may divide the extended area having the third size into an upper area and a lower area, and display different information related to the event in the divided areas. According to an embodiment, when the third size is larger than or equal to a set value and the event-related application includes a function of displaying an image and/or a video, the electronic device may divide the extended area having the third size into an upper area and a lower area, display a user interface capable of controlling content and/or at least a portion of content related to the event in one area, and display an image and/or video (e.g., a visual interface (VI)) of the application in the other area.

[0123] According to an embodiment, if the flexible display is configured to display a widget when its size is the third size or more, the electronic device may display a widget in one of the divided areas and display at least a portion of content related to the event, a user interface capable of controlling content and/or an image (or video) of the application in the other area.

[0124] According to an embodiment, an operation of extending the flexible display based on a user input and an operation of dividing the screen according to the size of the extended area are described in more detail with reference to FIGS. **10**, **11**A, and **11**B.
[0125] According to an embodiment, the electronic device may display a live view obtained by a camera (e.g., the camera module **180** of FIG. **1**) in the extended area based on an event being the execution of the camera application.
[0126] According to an embodiment, if a user input for capturing is received, the electronic device may obtain an image (or video) having the same aspect ratio as the extended area.
[0127] According to an embodiment, an operation of obtaining an image (or video) having the same aspect ratio as the aspect ratio of the extended area of the flexible display is described with reference to FIG. **12**.
[0128] According to an embodiment, the electronic device may display a preset widget in at least a portion of the extended area of the flexible display. According to an embodiment, when the flexible display is extended, the displayed widget may be set by a user input.
[0129] According to an embodiment, an operation of displaying a widget in an extended area of the flexible display is described below in more detail with reference to FIGS. **13**A and **13**B.
[0130] FIG. **7** is a view illustrating a use example when an electronic device is mounted on a flip cover according to an embodiment. For example, FIG. **7**(*a*) illustrates a state in which the flip cover is opened, FIG. **7**(*b*) illustrates a state in which the flip cover is partially closed, and FIG. **7**(*c*) illustrates a state in which the flexible display is extended in a state in which the flip cover is closed.
[0131] According to an embodiment, at least a portion of the housing of the electronic device **101** (e.g., the electronic device **101** of FIG. **1**) may be fastened to the flip cover **710**. According to an embodiment, the flip cover **710** may include a cover covering the flexible display (e.g., the display module **160** of FIG. **1**) of the electronic device **101**. According to an embodiment, a transparent material may be disposed in at least a partial area of the cover of the flip cover **710**. According to an embodiment, a card receiving portion may be disposed in at least a partial area of the flip cover **710**.
[0132] According to an embodiment, as illustrated in FIG. **7**(*a*), in the opened state of the flip cover **710**, the flexible display is not covered, and even when the flexible display is extended, not only the extended area but also the existing area may be used without disturbance.
[0133] According to an embodiment, as illustrated in FIG. **7**(*b*), in the at least partially closed state in which the cover of the flip cover **710** is rotated to cover at least a portion of the flexible display of the electronic device **101**, the electronic device **101** may detect that the flip cover **710** is at least partially closed through at least one sensor (e.g., the sensor module **176** of FIG. **1**).
[0134] According to an embodiment, as illustrated in FIG. **7**(*c*), in a case where the flip cover **710** is closed in a state in which the flexible display of the electronic device **101** is slid in, if an event occurs, the electronic device **101** may extend the flexible display and display information related to the event in the extended area outside the flip cover **710**.
[0135] As such, event notification and user interaction are possible even when the flip cover is in the closed state.
[0136] FIG. **8**A is a view illustrating an operation of extending a flexible display of an electronic device in a vertical direction in a state in which at least a portion thereof is covered according to an embodiment.
[0137] FIG. **8**B is a view illustrating an operation of extending a flexible display of an electronic device in a horizontal direction in a state in which at least a portion thereof is covered according to an embodiment.
[0138] Referring to FIGS. **8**A and **8**B, if an event occurs in a state in which the flip cover **710** is closed, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may extend the flexible display (e.g., the display module **160** of FIG. **1**) to a first size **810** and

820 (e.g., 30% of the maximum extendable size), a second size **811** and **821** (e.g., 50% of the maximum extendable size), and/or a third size **812** and **822** (e.g., the maximum extendable size) larger than the second size **811** and **821**.

[0139] According to an embodiment, the electronic device may extend the flexible display to a size set for each application in which an event occurs.

[0140] According to an embodiment, the electronic device may gradually or continuously extend the flexible display to the first size **810** and **820**, the second size **811** and **821**, or the third size **812** and **822**, according to a user input. For example, in a case where the flexible display is extended to the first size **810** or **820** based on an occurrence of an event, if a user input (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received, the electronic device may extend the flexible display to the second size **811** and **821**.

[0141] According to an embodiment, if a user input for extending the flexible display is received in a state in which the flexible display is extended to the second size **811** and **821**, the electronic device may extend the flexible display to the third size **821** and **831**.

[0142] Although three extension sizes of the flexible display are illustrated and described in FIGS. **8**A and **8**B, according to an embodiment, there may be two or less, or four or more extension sizes of the flexible display.

[0143] According to an embodiment, when the flexible display is extended to the first size **810** and **820** based on the occurrence of an event, if the user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is maintained (e.g., long press or long touch), the electronic device may continuously extend the flexible display until the user input is completed.

[0144] According to an embodiment, vertical extension of the flexible display is illustrated and described in FIGS. **9**A to **13**B, but the operations of the electronic device illustrated in FIGS. **9**A to **13**B are applicable even when the flexible display is extended in a horizontal direction.

[0145] FIG. **9**A is a view illustrating an extending operation based on an application where an event occurs, in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment. For example, FIG. **9**A illustrates a case where an event occurs in a music player application.

[0146] Referring to FIG. **9**A, if an event occurs in a state in which the flip cover **710** is closed, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may extend the flexible display (e.g., the display module **160** of FIG. **1**) by the first size **910** and display information **920** about the event corresponding to the first size **910** in which the event occurs in the extended area having the first size **910**. According to an embodiment, the information **920** about the event displayed in the extended area having the first size **910** may be a color corresponding to the application in which the event has occurred. According to an embodiment, the color corresponding to the application may be a main color (e.g., green) of the icon of the application.

[0147] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the first size **910**, the electronic device may extend the flexible display to the second size **911** corresponding to the application in which the event has occurred.

[0148] According to an embodiment, the electronic device may display information **921** about the event corresponding to the second size **911** on the flexible display extended to the second size **911**. According to an embodiment, the information **921** about the event corresponding to the second size **911** may include a summary for content and/or a user interface (e.g., information about the music being played (song title and/or singer), playback/pause button, previous song button, and next song button) for content control.

[0149] FIG. **9**B is a view illustrating an extending operation based on an application where an event occurs, in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment. For example, FIG. **9**B illustrates a case where an event occurs in a video player application.

[0150] Referring to FIG. **9**B, if an event occurs in a state in which the flip cover **710** is closed, the electronic device may extend the flexible display by the first size **910** and display information **930** about the event corresponding to the first size **910** in the extended area having the first size **910**. According to an embodiment, the information **930** about the application displayed in the extended area having the first size **910** may be a color corresponding to the application in which an event has occurred. According to an embodiment, the color corresponding to the application may be a main color (e.g., red) of the icon of the application.

[0151] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the first size **910**, the electronic device may extend the flexible display to the third size **912** corresponding to the application in which the event has occurred.

[0152] According to an embodiment, the electronic device may display information **931** about the event corresponding to the third size **912** on the flexible display extended to the third size **912**. According to an embodiment, the information **931** about the event corresponding to the third size **912** may include content (e.g., video) about the content.

[0153] As such, in a state in which the flip cover **710** is closed, a different color may be displayed and the flexible display may be extended to a different size according to the application in which the event occurs.

[0154] FIG. **10** is a view illustrating a stepwise extending operation when an event occurs in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.

[0155] Referring to FIG. **10**, if an event occurs in a state in which the flip cover **710** is closed, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may extend the flexible display (e.g., the display module **160** of FIG. **1**) by the first size and display information **1011** about the event corresponding to the first size **1010** in the extended area having the first size **1010**. According to an embodiment, the information **1011** about the event displayed in the extended area having the first size **1010** may be a color corresponding to the application in which the event has occurred. According to an embodiment, the color corresponding to the application may be a main color (e.g., yellow) of the icon of the application.

[0156] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the first size **1010**, the electronic device may extend the flexible display to the second size **1020** corresponding to the application in which the event has occurred.

[0157] According to an embodiment, the electronic device may display information **1021** about the event corresponding to the second size **1020** on the flexible display extended to the second size **1020**. According to an embodiment, the information **1021** about the event corresponding to the second size **1020** may include a summary (e.g., a new message arrival notification) of the content and/or a user interface for controlling the content.

[0158] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the second size **1020**, the electronic device may extend the flexible display to the third size **1030** corresponding to the application in which the event has occurred.

[0159] According to an embodiment, the electronic device may display information **1031** about the event corresponding to the third size **1030** on the flexible display extended to the third size **1030**. According to an embodiment, the information **1031** about the event corresponding to the third size

1030 may include content (e.g., message content) about the content.

[0160] Although FIG. **10** illustrates that the flexible display is extended stepwise and information corresponding to the size of the extended area is displayed, according to an embodiment, when the user input is continuous (e.g., a long press on a physical key or a long touch on the flexible display), the electronic device may extend the flexible display until the user input is completed and display information about the event corresponding to the size of the extended flexible display.

[0161] As such, notification of the occurrence of an event and user interaction are possible even when the flip cover is closed.

[0162] FIG. **11**A is a view illustrating an operation in which a screen configuration is varied during stepwise extension in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.

[0163] FIG. **11**B is a view illustrating an operation in which a screen configuration is varied during stepwise extension in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.

[0164] Referring to FIG. **11**A, if an event occurs in a state in which the flip cover **710** is closed, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may extend the flexible display (e.g., the display module **160** of FIG. **1**) by the first size and display information **1120** about the event corresponding to the first size **1110** in the extended area having the first size **1110**. According to an embodiment, the information **1120** about the event displayed in the extended area having the first size **1110** may be a color corresponding to the application in which the event has occurred. According to an embodiment, the color corresponding to the application may be a main color of the application icon.

[0165] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the first size **1110**, the electronic device may extend the flexible display to the second size **1111** corresponding to the application in which the event has occurred.

[0166] According to an embodiment, the electronic device may display information **1130** about the event corresponding to the second size **1111** on the flexible display extended to the second size **1111**. According to an embodiment, the information **1130** about the event corresponding to the second size **1111** may include a summary of content and/or a user interface for content control. For example, as illustrated in FIG. **11**B, information **1131** about the event corresponding to the second size **1111** may include information about the song being played (song title and/or singer), a play/pause button, a previous song button, and/or a next song button.

[0167] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the second size **1111**, the electronic device may extend the flexible display to the third size **1112** corresponding to the application in which the event has occurred.

[0168] According to an embodiment, the electronic device may divide the extended area having the third size **1112** into a left area and a right area, display a summary of content related to the event and/or a user interface (UI) capable of controlling content in one area **1140**, and display an image and/or video (e.g., a visual interface (VI)) of the application in the other area **1141**.

[0169] For example, as illustrated in FIG. **11**B, the electronic device may display information about the song being played (song title and/or singer), playback/pause button, previous song button, and/or next song button) in one area **1142** of the divided left and right areas of the extended area having the third size **1112** of the flexible display, and display an image or video related to the song being played in the other area **1143**.

[0170] According to an embodiment, if a user input for extending the flexible display (e.g., pressing a physical key for extending the flexible display or touching the extended area) is received in a state in which the flexible display is extended to the third size **1112**, the electronic device may extend the flexible display to a fourth size **1113** corresponding to the application in which the event has occurred.

[0171] According to an embodiment, the electronic device may divide the extended area having the fourth size **1113** into an upper area and a lower area and display different information related to the event in the divided areas. A summary of content related to the event and/or a user interface (UI) capable of controlling content may be displayed in one area **1150**, and an image and/or video (e.g., a visual interface (VI)) of the application may be displayed in the other area **1151**.

[0172] For example, as illustrated in FIG. **11**B, the electronic device may display information about the song being played (song title and/or singer), playback/pause button, previous song button, and/or next song button) in one area **1152** of the divided upper and lower areas of the extended area having the fourth size **1113** of the flexible display, and display an image or video related to the song being played in the other area **1153**.

[0173] By differently arranging the divided areas based on the extension size of the flexible display as described above, information including an image and/or a video may be provided more efficiently.

[0174] FIG. **12** is a view illustrating an operation of obtaining an image based on an aspect ratio of an extended area when a camera application is executed in a state in which at least a portion thereof a flexible display of an electronic device is covered according to an embodiment.

[0175] Referring to FIG. **12**, if an event of executing a camera application occurs in a state in which the flip cover **710** is closed, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may extend the flexible display (e.g., the display module **160** of FIG. **1**) by the first size and display a live view obtained by the camera (e.g., the camera module **180** of FIG. **1**) in the extended area having the first size. According to an embodiment, the camera may be a rear camera or a front camera of the electronic device.

[0176] According to an embodiment, if a user input for capturing is received in a state in which a live view is displayed on the flexible display extended to the first size, the electronic device may obtain an image **1210** (or video) having the same aspect ratio as the first size.

[0177] According to an embodiment, when the flexible display is extended by a second size larger than the first size based on receiving a user input (e.g., pressing a physical key for extending the flexible display or touching a user interface touch of the flexible display), the electronic device may display a live view obtained by the camera in the extended area having the second size.

[0178] According to an embodiment, if a user input for capturing is received in a state in which a live view is displayed on the flexible display extended to the second size, the electronic device may obtain an image **1220** (or video) having the same aspect ratio as the second size.

[0179] According to an embodiment, when the flexible display is extended by a third size larger than the second size based on receiving a user input (e.g., pressing a physical key for extending the flexible display or touching a user interface of the flexible display), the electronic device may display a live view obtained by the camera in the extended area having the third size.

[0180] According to an embodiment, if a user input for capturing is received in a state in which a live view is displayed on the flexible display extended to the third size, the electronic device may obtain an image **1230** (or video) having the same aspect ratio as the third size.

[0181] FIG. **13**A is a view illustrating a configuration screen for receiving a user input for setting to display a widget when a flexible display is extended based on an occurrence of an event in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.

[0182] Referring to FIG. **13**A, the electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may display a setting screen **1310** for setting a widget to be displayed in the extended area when the flexible display (e.g., the display module **160** of FIG. **1**) is extended in a state in which the flip cover (e.g., the flip cover **710** of FIG. **7**) is closed. According to an

embodiment, a user input for setting the widget **1311** to be
[0183] displayed in the extended area may be received through the setting screen **1310**. According to an embodiment, the application to be displayed as the widget, the type of the widget, the position where the widget is to be displayed, and/or the size of the widget may be set through the setting screen **1310**.
[0184] According to an embodiment, a different widget may be set for each size of the extended area through the setting screen **1310**.
[0185] FIG. **13**B is a view illustrating a widget displayed based on a degree of extension of a flexible display in a state in which at least a portion of a flexible display of an electronic device is covered according to an embodiment.
[0186] Referring to FIG. **13**B, the electronic device may extend the flexible display by the first size **1320** based on the occurrence of an event in a state in which the flip cover (e.g., the flip cover **710** of FIG. **7**) is closed, and display information about the event corresponding to the first size **1320** in the extended area having the first size **1320**.
[0187] According to an embodiment, based on the setting of the electronic device, the electronic device may divide the extended area having the first size **1320** into a left area and a right area, display a widget **1330** disposed by a user input in one area **1330**, and display a summary of content related to the event and/or a user interface capable of controlling content in the other area **1331**.
[0188] According to an embodiment, based on receiving a user input for extending the flexible display in a state in which the flexible display is extended by the first size **1320**, the flexible display may be extended by a second size **1321** larger than the first size **1320**.
[0189] According to an embodiment, the electronic device may further display a second widget **1332** disposed by a user input in the area extended further than the first size of the extended area having the second size **1321**.
[0190] As described above, even when the flip cover is closed, interaction with the user may be possible through extension of the flexible display.
[0191] According to an embodiment, an electronic device may comprise a housing, at least one sensor, a flexible display at least partially exposed to an outside through the housing, and at least one processor operatively connected, directly or indirectly, to the at least one sensor and the flexible display.
[0192] According to an embodiment, the at least one processor may identify whether at least a portion of the flexible display is covered through the at least one sensor in a state in which the flexible display slides in with respect to the housing.
[0193] According to an embodiment, the at least one processor may perform a slide-out operation of the flexible display based on an occurrence of an event in the state in which the at least a portion of the flexible display is covered.
[0194] According to an embodiment, the at least one processor may display information related to the event in an area extended by the slide-out operation.
[0195] According to an embodiment, the at least one processor may identify that the at least a portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.
[0196] According to an embodiment, the at least one processor may perform the slide-out operation of the flexible display to extend the flexible display by a first size based on the occurrence of the event.
[0197] According to an embodiment, the at least one processor may display a color corresponding to an application corresponding to the event in the extended area having the first size.
[0198] According to an embodiment, the at least one processor may perform the slide-out operation of the flexible display to extend the flexible display by a second size larger than the first size based on receiving a user input in a state in which the flexible display is extended by the first size.
[0199] According to an embodiment, the at least one processor may display information related to

the event corresponding to the second size in the extended area having the second size.
[0200] According to an embodiment, the at least one processor may divide the extended area having the second size into a left area and a right area.
[0201] According to an embodiment, the at least one processor may display different information related to the event in the divided areas.
[0202] According to an embodiment, the at least one processor may perform the slide-out operation of the flexible display to extend the flexible display by a third size larger than the second size based on receiving a user input in a state in which the flexible display is extended by the second size.
[0203] According to an embodiment, the at least one processor may display information related to the event corresponding to the third size in the extended area having the third size.
[0204] According to an embodiment, the at least one processor may divide the extended area having the third size into an upper area and a lower area.
[0205] According to an embodiment, the at least one processor may display different information related to the event in the divided areas.
[0206] According to an embodiment, the at least one processor may determine an extension size of the flexible display based on the application corresponding to the event.
[0207] According to an embodiment, the electronic device may further comprise a camera.
[0208] According to an embodiment, the at least one processor may display a live view obtained by the camera in the extended area based on the event being execution of a camera application.
[0209] According to an embodiment, the at least one processor may obtain an image having the same aspect ratio as an aspect ratio of the extended area if a user input for capturing is received.
[0210] According to an embodiment, the at least one processor may display a preset widget in at least a portion of the extended area.
[0211] According to an embodiment, a method for controlling an electronic device may comprise identifying whether at least a portion of a flexible display is covered through at least one sensor in a state in which the flexible display slides in with respect to a housing.
[0212] According to an embodiment, the method for controlling the electronic device may comprise sliding out the flexible display based on an occurrence of an event in the state in which the at least a portion of the flexible display is covered.
[0213] According to an embodiment, the method for controlling the electronic device may comprise displaying information related to the event in an area extended by the slide-out operation.
[0214] According to an embodiment, identifying that the at least a portion of the flexible display is covered may identify that the at least a portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.
[0215] According to an embodiment, sliding out the flexible display may slide out the flexible display to extend the flexible display by a first size based on the occurrence of the event.
[0216] According to an embodiment, displaying the information related to the event may display a color corresponding to an application corresponding to the event in the extended area having the first size.
[0217] According to an embodiment, the method for controlling the electronic device may further comprise sliding out the flexible display to extend the flexible display by a second size larger than the first size based on receiving a user input in a state in which the flexible display is extended by the first size.
[0218] According to an embodiment, the method for controlling the electronic device may further comprise displaying information related to the event corresponding to the second size in the extended area having the second size.
[0219] According to an embodiment, displaying the information related to the event corresponding to the second size may divide the extended area having the second size into a left area and a right area and display different information related to the event in the divided areas.
[0220] According to an embodiment, the method for controlling the electronic device may further

comprise sliding out the flexible display to extend the flexible display by a third size larger than the second size based on receiving a user input in a state in which the flexible display is extended by the second size.

[0221] According to an embodiment, the method for controlling the electronic device may further comprise displaying information related to the event corresponding to the third size in the extended area having the third size.

[0222] According to an embodiment, displaying the information related to the event corresponding to the third size may divide the extended area having the third size into an upper area and a lower area and display different information related to the event in the divided areas.

[0223] According to an embodiment, sliding out the flexible display may further include determining an extension size of the flexible display based on the application corresponding to the event.

[0224] According to an embodiment, displaying the information related to the event may display a live view obtained by the camera in the extended area based on the event being execution of a camera application.

[0225] According to an embodiment, displaying the information related to the event may further include obtaining an image having the same aspect ratio as an aspect ratio of the extended area if a user input for capturing is received.

[0226] According to an embodiment, displaying the information related to the event may display a preset widget in at least a portion of the extended area.

[0227] According to an embodiment, in a non-transitory, computer-readable recording medium storing one or more programs, the one or more programs may comprise instructions that, according to an embodiment, enable the at least one processor of an electronic device to identify whether at least a portion of the flexible display is covered through the at least one sensor in a state in which the flexible display slides in with respect to the housing.

[0228] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to perform a slide-out operation of the flexible display based on an occurrence of an event in the state in which the at least a portion of the flexible display is covered.

[0229] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display information related to the event in an area extended by the slide-out operation.

[0230] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to identify that the at least a portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.

[0231] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to perform the slide-out operation of the flexible display to extend the flexible display by a first size based on the occurrence of the event.

[0232] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display a color corresponding to an application corresponding to the event in the extended area having the first size.

[0233] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to perform the slide-out operation of the flexible display to extend the flexible display by a second size larger than the first size based on receiving a user input in a state in which the flexible display is extended by the first size.

[0234] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display information related to the event corresponding to the second size in the extended area having the second size.

[0235] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to divide the extended area having the second size into a left area and a right area.

[0236] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display different information related to the event in the divided areas.
[0237] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to perform the slide-out operation of the flexible display to extend the flexible display by a third size larger than the second size based on receiving a user input in a state in which the flexible display is extended by the second size.
[0238] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display information related to the event corresponding to the third size in the extended area having the third size.
[0239] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to divide the extended area having the third size into an upper area and a lower area.
[0240] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display different information related to the event in the divided areas.
[0241] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to determine an extension size of the flexible display based on the application corresponding to the event.
[0242] According to an embodiment, the electronic device may further comprise a camera.
[0243] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display a live view obtained by the camera in the extended area based on the event being execution of a camera application.
[0244] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to obtain an image having the same aspect ratio as an aspect ratio of the extended area if a user input for capturing is received.
[0245] According to an embodiment, the one or more programs may comprise instructions that enable the electronic device to display a preset widget in at least a portion of the extended area.
[0246] The electronic device according to an embodiment may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
[0247] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element. Thus, “connected” as used herein covers both direct and indirect connections.

[0248] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC). Thus, each “module” herein may comprise circuitry.

[0249] An embodiment of the disclosure may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The storage medium readable by the machine may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0250] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0251] According to an embodiment, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to an embodiment, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device, comprising: a housing; at least one sensor; a flexible display at least partially visible to an outside; memory storing at least one instruction; and at least one processor, comprising processing circuitry, operatively connected to the at least one sensor, the flexible display and the memory, wherein the at least one processor, individually or collectively, is configured to execute the at least one instruction and to cause the electronic device to: identify whether a portion of the flexible display is covered through the at least one sensor in a state in which the flexible display slides in with respect to the housing; perform a slide-out operation of the

flexible display based on an occurrence of an event in a state in which the portion of the flexible display is covered; and display information related to the event in an area exposed by the slide-out operation, the exposed area corresponding to the covered portion.
2. The electronic device of claim 1, wherein the at least one processor is individually or collectively configured to cause the electronic device to identify that the portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.
3. The electronic device of claim 1, wherein the at least one processor is individually or collectively configured to cause the electronic device to: perform the slide-out operation of the flexible display to extend a visible area of the flexible display by a first size based on the occurrence of the event; and displays a color corresponding to an application corresponding to the event in the exposed area having the first size.
4. The electronic device of claim 3, wherein the at least one processor is individually or collectively configured to cause the electronic device to: perform the slide-out operation of the flexible display to extend the visible area of the flexible display by a second size larger than the first size based on receiving a user input in a state in which the visible area of the flexible display is extended by the first size; and display information related to the event corresponding to the second size in the exposed area having the second size.
5. The electronic device of claim 4, wherein the at least one processor is individually or collectively configured to cause the electronic device to divide the exposed area having the second size into a left area and a right area and displays different information related to the event in the divided areas.
6. The electronic device of claim 4, wherein the at least one processor is individually or collectively configured to cause the electronic device to: perform the slide-out operation of the flexible display to extend the visible area of the flexible display by a third size larger than the second size based on receiving a user input in a state in which the visible area of the flexible display is extended by the second size; and display information related to the event corresponding to the third size in the exposed area having the third size.
7. The electronic device of claim 6, wherein the at least one processor is individually or collectively configured to cause the electronic device to divide the exposed area having the third size into an upper area and a lower area and display different information related to the event in the divided areas.
8. The electronic device of claim 1, wherein the at least one processor is individually or collectively configured to cause the electronic device to determine an extension size of the visible area of the flexible display based on the application corresponding to the event.
9. The electronic device of claim 1, further comprising a camera, wherein the at least one processor is individually or collectively configured to: display a live view obtained by the camera in the exposed area based on the event being execution of a camera application; and obtain an image having the same aspect ratio as an aspect ratio of the exposed area if a user input for capturing is received.
10. The electronic device of claim 1, wherein the at least one processor is individually or collectively configured to cause the electronic device to display a preset widget in at least a portion of the exposed area.
11. A method for controlling an electronic device, the method comprising: identifying whether a portion of a flexible display is covered through at least one sensor in a state in which the flexible display slides in with respect to a housing; sliding out the flexible display based on an occurrence of an event in the state in which the portion of the flexible display is covered; and displaying information related to the event in an area exposed by the slide-out operation, the exposed area corresponding to the covered portion.
12. The method of claim 11, wherein identifying that the portion of the flexible display is covered identifies that the at least a portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.

13. The method of claim 11, wherein sliding out the flexible display slides out the flexible display to extend a visible area of the flexible display by a first size based on the occurrence of the event, and wherein displaying the information related to the event displays a color corresponding to an application corresponding to the event in the exposed area having the first size.
14. The method of claim 13, further comprising: sliding out the flexible display to extend the visible area of the flexible display by a second size larger than the first size based on receiving a user input in a state in which the visible area of the flexible display is extended by the first size; and displaying information related to the event corresponding to the second size in the exposed area having the second size.
15. The method of claim 14, wherein displaying the information related to the event corresponding to the second size divides the exposed area having the second size into a left area and a right area and displays different information related to the event in the divided areas.
16. A non-transitory computer-readable recording medium storing at least one instruction which, when executed by at least one processor, individually or collectively, of an electronic device, causes the electronic device to perform operations including: identifying whether a portion of a flexible display is covered through at least one sensor in a state in which the flexible display slides in with respect to a housing; sliding out the flexible display based on an occurrence of an event in the state in which the portion of the flexible display is covered; and displaying information related to the event in an area exposed by the slide-out operation, the exposed area corresponding to the covered portion.
17. The non-transitory computer-readable recording medium of claim 16, wherein identifying that the portion of the flexible display is covered identifies that the at least a portion of the flexible display is covered based on identifying that a flip cover is closed through the at least one sensor.
18. The non-transitory computer-readable recording medium of claim 16, wherein sliding out the flexible display slides out the flexible display to extend a visible area of the flexible display by a first size based on the occurrence of the event, and wherein displaying the information related to the event displays a color corresponding to an application corresponding to the event in the exposed area having the first size.
19. The non-transitory computer-readable recording medium of claim 18, the operations further comprising: sliding out the flexible display to extend the visible area of the flexible display by a second size larger than the first size based on receiving a user input in a state in which the visible area of the flexible display is extended by the first size; and displaying information related to the event corresponding to the second size in the exposed area having the second size.
20. The non-transitory computer-readable recording medium of claim 19, wherein displaying the information related to the event corresponding to the second size divides the exposed area having the second size into a left area and a right area and displays different information related to the event in the divided areas.